

HERWIJNEN, Netherlands

WMO: 063560

Lat: 51.8581N

Lon: 5.1452E

Elev: 3

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.9	(c) 23.8	(d) 14.6	(e) 11.5	(f) 23.0	(g) 19.1	(h) 14.4	(i) 24.8	(j) 31.9	(k) 49.2	(l) 28.8	(m) 48.2	(n) 6.0	(o) 80	(p) 0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	83.3	68.5	79.4	66.9	76.1	65.1	70.3	80.3	68.4	77.1	66.6	73.8	7.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	98.5	74.6	65.2	93.0	72.2	63.7	88.4	70.4	34.2	80.3	32.6	77.4	31.2	74.1	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.1	(b) 21.6	(c) 18.8	(e) 14.2	(f) 91.0	(g) 8.5	(h) 3.1	(i) 8.0	(j) 93.3	(k) 3.0	(l) 95.1	(m) -1.7	(n) 96.8	(o) -7.9	(p) 99.1		
			DB													
			WB													
				13.5	74.9	8.3	2.0	7.5	76.4	2.6	77.6	-2.0	78.8	-8.1	80.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	37.9	39.0	43.3	49.2	56.0	61.0	64.4	63.5	58.8	52.0	44.6	39.3		
	DBStd	11.31	7.31	6.93	6.12	6.54	6.35	5.52	5.44	4.69	4.96	5.91	6.11	7.58		
	HDD50	1576	377	309	218	93	16	0	0	0	1	47	179	336		
	HDD65	5318	841	727	673	474	287	145	79	85	195	403	612	797		
	CDD50	1874	1	2	10	69	202	330	445	419	264	110	18	4		
	CDD65	138	0	0	0	1	8	24	59	38	8	0	0	0		
	CDH74	1215	0	0	0	14	88	243	497	298	72	3	0	0		
	CDH80	296	0	0	0	0	11	59	141	73	12	0	0	0		
(14) Wind	WSAvg	8.9	10.8	10.4	9.7	8.5	8.4	7.9	7.8	7.3	7.3	8.7	9.3	10.5		
(15) Precipitation	PrecAvg	33.0	2.8	2.5	2.1	1.8	2.4	2.8	3.4	3.2	2.8	2.9	3.1	3.2		
	PrecMax	40.9	5.0	5.1	3.9	3.9	5.1	5.2	7.0	7.4	6.9	7.3	5.4	6.5		
	PrecMin	23.2	0.2	0.6	0.5	0.0	0.8	0.6	1.3	0.6	0.4	0.9	0.4	0.8		
	PrecStd	4.6	1.3	1.1	1.1	0.9	1.1	1.3	1.6	1.8	1.5	1.4	1.2	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	57.3	64.8	76.2	81.3	86.5	89.8	87.3	81.0	71.7	61.5	56.2		
		MCWB	52.3	50.9	55.2	60.4	66.5	70.2	70.2	70.9	67.5	63.3	57.4	53.3		
	2%	DB	52.2	53.0	59.8	70.0	76.1	80.3	83.5	81.2	75.6	65.9	58.1	54.0		
		MCWB	50.1	49.3	52.2	57.8	62.1	66.8	68.6	68.5	65.8	60.9	55.3	51.7		
	5%	DB	49.9	50.9	55.7	65.1	72.5	75.9	79.2	76.9	71.5	62.8	55.8	52.0		
		MCWB	47.9	48.0	50.3	55.6	60.6	65.0	67.1	66.5	63.7	58.7	53.2	49.9		
	10%	DB	47.7	48.5	52.8	60.8	68.2	72.1	75.5	73.0	67.7	60.6	53.4	49.6		
		MCWB	45.8	46.2	48.8	53.3	58.9	63.0	65.6	64.4	61.4	57.2	51.2	47.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	52.1	55.9	62.6	68.2	72.0	72.8	72.7	69.3	64.1	58.2	54.0		
		MCDB	54.4	55.1	61.8	74.3	78.9	83.2	83.7	83.0	78.4	69.5	60.1	55.5		
	2%	WB	50.4	50.1	53.4	58.9	64.8	68.5	70.3	69.8	66.7	61.4	55.6	52.1		
		MCDB	52.1	52.3	58.3	67.9	72.7	77.5	80.7	78.8	74.0	65.4	57.8	53.8		
	5%	WB	48.1	48.3	51.2	56.4	61.8	66.2	68.4	67.6	64.6	59.5	53.5	50.2		
		MCDB	49.7	50.4	54.8	64.0	69.8	74.0	77.2	75.0	69.7	62.5	55.5	51.9		
	10%	WB	46.1	46.4	49.2	54.2	59.6	64.0	66.4	65.6	62.6	57.6	51.4	47.9		
		MCDB	47.7	48.6	52.5	59.8	66.8	70.9	73.6	71.3	67.1	60.1	53.2	49.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	10.5	14.0	17.7	17.4	17.4	16.5	16.7	15.9	12.8	10.0	8.4		
		MCDBR	9.0	12.8	19.7	24.4	23.8	23.9	23.6	22.8	21.0	14.8	10.9	8.5		
		MCWBR	8.2	9.9	13.5	14.3	12.5	12.8	11.1	11.8	12.0	10.1	9.1	7.7		
	5% WB	MCDBR	9.1	11.3	17.5	21.9	21.1	21.1	21.0	20.6	18.7	14.3	10.7	8.8		
		MCWBR	8.5	9.2	12.6	13.5	12.0	12.2	10.9	11.7	11.4	10.4	9.3	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.367	0.401	0.427	0.434	0.429	0.439	0.429	0.403	0.380	0.356	0.339		
	taud		2.299	2.269	2.184	2.157	2.187	2.220	2.201	2.249	2.307	2.334	2.311	2.284		
	Ebn at Noon		203	233	247	256	260	262	257	253	245	225	195	181		
	Edh at Noon		21	29	37	43	43	43	43	39	33	26	21	18		
(44) All-Sky Solar Radiation	RadAvg		229	446	824	1328	1606	1724	1625	1374	1003	581	266	173		
	RadStd		26	84	107	136	119	127	183	116	89	48	26	21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page